Paweł Duda 422637
Mieszko Karski 419972
Bartłomiej Łukaszczyk 419866

Mechatronic engineering, group 1

Mechatronic design

Intelligent bird feeder



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1. Assumptions

1.1. Project description

The goal of the project is to design an autonomous, low-power mechatronic bird feeder that uses electronic weight sensing to selectively feed wildlife.
The idea is to design a birdhouse with a weight-activated door mechanism, which triggers when a large intruder (e.g. pigeon, squirrel) tries to enter the feeder but does not obstruct smaller birds (e.g. sparrows).
The design should be very energy efficient.
It will record the bird landings to an SD card.

1.2. State of the art

Currently available products

- **Standard mechanical solutions** (e.g. Squirrel Buster)
  Purely mechanical designs, which use a calibrated internal spring. When a heavy animal lands its weight pulls a metal shroud downwards, physically blocking the seed. Their downside is they cannot log any data and if weight needs to be adjusted, the entire feeder has to be taken apart.

- **Motorised spinners** (e.g. Yankee Flipper)
  A mechanical weight switch that triggers an internal DC motor that spins, tossing the squirrel off. Its drawbacks are that it lacks data logging, weight selectivity is difficult and squirrels tend to learn how to block the motor.

- **AI vision feeders** (e.g. Bird Buddy)
  They use cameras and cloud AI to identify birds, providing specialised data. However, they are very expensive, require WiFi, drain batteries quickly and cause privacy concerns.

What the project does differently

The project's main strength is being able to log data without requiring WiFi or Bluetooth connection, making it much more energy efficient, as seen in section 3.3, the design could theoretically work indefinitely without requiring battery change. It is also quite robust to squirrels and pigeons outsmarting the design, as all the electrical components are hidden within a secure box, making it impossible to reach. Moreover, the software implemented has a built in tare function, which removes the problem of weight overload when seed is added or during heavy snowfall.

1.3. Target group

The project is focused on three main target groups:

- **Bird enthusiasts**
  Hobbyists, who want to track the feeding habits, visit frequencies and approximate species of their local birds (inferred via weight data) over the seasons.

- **Gardeners**
  People desperate for a highly reliable, adjustable squirrel and pigeon deterrent that cannot be easily outsmarted.

- **Privacy conscious consumers**
  Users who want smart, data-logging garden tech but do not want an internet-connected camera recording their property 24/7.

1.4. Morphological analysis

Criteria

The morphological analysis was done according to three main criteria, marked in the table below (Table 1.1) in their corresponding colours:

- **Functionality:** **1-10**
  How well does the part fulfill its requirement?

  1 – The requirement is not fulfilled
  10 – Fulfills the requirement perfectly

- **Energy consumption:** **1-10**
  How energy efficient is the part?

  1 – The part consumes a lot of energy
  10 – The part is very energy efficient

- **Cost and simplicity:** **1-10**
  How expensive is the part and how difficult would it be to implement it?

  1 – The part is costly / difficult to implement
  10 – The part is cheap / simple to implement

Table

Aspects/functions	Option 1	Option 2	Option 3	Option 4
Weight measurement	9|7|8 Load cell	3|10|5 Calibrated spring	3|9|5 Pressure pad	
Actuator type	9|8|8 Servo motor	6|7|6 DC motor	8|3|4 Stepper motor	
Deterrent mechanism	4|8|7 Top-down door	8|8|7 Bottom-up door	8|7|6 Rotary shield	5|5|4 Lowered perch
Data storing	9|9|9 SD card	2|10|5 RAM microcontroller	6|7|7 Bluetooth	8|1|2 Wi-Fi
Power source	9|9|8 Solar panel + Li-Ion	5|2|8 AA batteries	3|8|1 Wind turbine	
Sturdy design	10|-|3 Aluminium	9|-|6 PC	5|-|9 3D printed ABS	7|-|8 Hardwood

Table 1.1 – Morphological analysis table (final choices for the prototype are marked in yellow)

Detailed description

- **Weight measurement**
  - Load cell 9|**7**|**8** (Score: 24/30)
    Highly accurate, provides precise digital data for logging and is relatively cheap to interface with an HX711 amplifier. It requires software auto-taring to counter temperature drift but it is the definitive best choice.

  - Calibrated spring 3|**10**|**5** (Score: 18/30)
    Extremely power efficient (zero power), but terrible for a data-logging system. To get electronic data from a spring, one would need complex optical encoders or multiple microswitches, defeating the simplicity.

  - Pressure pad 3|**9**|**5** (Score: 17/30)
    Cheap and low power, but highly non-linear and wildly inaccurate across different outdoor temperatures. It cannot reliably distinguish between a 40g bird and a 60g bird.

- **Actuator type**
  - Servo motor 9|**8**|**8** (Score: 25/30)
    Ideal for the project. It has built-in closed-loop control and only draws significant current whilst moving. Once the door is shut, power can be cut and the mechanical gearing holds it in place.

- DC motor 6|**8**|6 (Score: 20/30)
  Simple to drive, but it has no internal position tracking. One would need to add external limit switches to tell the microcontroller when the door is fully open or closed, adding mechanical complexity and points of failure.

- Stepper 8|**3**|4 (Score: 15/30)
  Excellent precision, but completely unnecessary for a simple door. They are heavy, expensive and notoriously power hungry as they often require constant current to hold their position.

- **Deterrent mechanism**
  - Top-down door 4|**8**|5 (Score: 17/30)
    Risk of working as a guillotine. Prone to trapping claws or beaks if the sensor is triggered accidentally. High mechanical risk.

  - Bottom-up door 8|**8**|7 (Score: 23/30)
    Much safer than a top-down door, as it pushes the animal up and away rather than pinning it. The only downside is the possible dust buildup over time could cause jamming.

  - Rotary shield 8|**7**|6 (Score: 21/30)
    A curved piece of plastic or metal that rotates over the seed ports. However, it would be difficult to implement while trying to avoid the risk of squirrels learning to pry it open.

  - Lowered perch 5|**5**|4 (Score: 14/30)
    Dropping the perch to dump the animal requires heavy actuation, drawing high current and the physical violence of the mechanism causes risk concerns.

- **Data storing**
  - Micro SD card 9|**9**|9 (Score: 27/30)
    Perfect for ecological data. It is non-volatile, consumes practically zero power when not writing and CSV files take up tiny amounts of space.

  - RAM Microcontroller 2|**10**|5 (Score: 17/30)
    Volatile memory. If the battery dies or the system is reset, all the data is instantly erased.

  - Bluetooth 6|**7**|7 (Score: 20/30)
    Good for transferring data to a phone without opening the feeder, but requires more programming and a higher standby power draw.

  - Wi-Fi 8|**1**|2 (Score: 11/30)
    Overkill for text data, requiring router connection and battery draining.

- **Power source**
  - Solar panel + Li-Ion 9|**9**|**8** (Score: 26/30)
    The gold standard. A 5V solar panel paired with an 18650 lithium-ion cell and a charging module will keep a sleep-optimised microcontroller running indefinitely (as calculated in section 3.3).

  - AA batteries 5|**2**|**8** (Score: 15/30)
    Requires the user to frequently dismantle the feeder to swap batteries. Not only irritating but also bad for the environment.

  - Wind turbine 3|**8**|**1** (Score: 12/30)
    Most bird feeders are placed in sheltered gardens or near trees. A micro-turbine would rarely catch enough consistent wind to charge the system and it adds unnecessary moving parts which can be damaged by the animals (or damaging to them).

- **Sturdy design**
  - Aluminium 10|-|**3** (Score: 13/20)
    Very resistant to weather and animal manipulations. It is, however, expensive to source and requires specialised tools to machine and assemble.

  - Polycarbonate 8|-|**6** (Score: 14/20)
    Highly durable and UV resistant. However, weather-proofing the PC tubes requires careful designing.

  - 3D printed ABS 5|-|**9** (Score: 14/20)
    It is very cheap and good for creating complex geometries, but standard ABS can suffer from layer splitting in fluctuating outdoor temperatures and humidities and a determined squirrel can chew through it.

  - Hardwood 7|-|**8** (Score: 15/20)
    A classic choice, offering an aesthetically pleasing design which is quite sturdy at the same time. It is also relatively cheap and only requires simple tools to shape and assemble.

1.5. Design assumptions

Final design assumptions concluded from morphological analysis (1.4):

- **Project:** The bulk of the project will be constructed out of hardwood, with some moving components, like the door, requiring reinforced 3D prints.
  Prototype: The prototype will be constructed out of plywood to lower the weight and save on costs. The door mechanism will be 3D printed only to show the proof of concept.

- **Project:** The project will implement a high load strain gauge connected to a bottom-up door mechanism on a servo motor. This will allow for accurate weight measurements and a safe but stern deterrent mechanism, prone to being outsmarted.
  Prototype: The prototype will mirror those assumptions, using a smaller load cell due to costs and a mock door mechanism only showing the proof of concept.

- **Project:** The data will be stored on a micro SD card, easily accessible via a sliding door incorporated into the model's lower body.
  Prototype: The prototype will also incorporate the SD card, however, the construction of the sliding door was skipped due to the prototype's open design. The SD card will be available to investigate at any point.

- **Project:** A solar panel in pair with a lithium-ion core will be implemented, powering a sleep-optimised microcontroller, in theory to work indefinitely.
  Prototype: The prototype will use a cheap Arduino Uno microcontroller, not an optimal choice but readily available and enough to show the proof of concept. It will be powered directly from the computer via a USB cable.

1.6. Design of the mechanical part

Design process

The mechanical design of the bird feeder focuses on structural durability, weather resistance and component integration to achieve the objectives of the system.

Structure: The main enclosure is planned as sealed housing constructed primarily out of hardwood for durability and low cost, while the prototype utilizes plywood to minimize weight during testing. The enclosure isolates the delicate electronic components (Arduino, HX711, RTC and SD card reader) inside an internal dry box, keeping them out of reach of pests as well as the elements.
Weight sensing: To support the weight-selective feeding strategy, the feeder is mounted on a cantilever beam which is connected to the strain-gauge. This

arrangement causes any weight that rests on the bird feeder to be transferred to the sensor.
The door mechanism: The selected door mechanism is a bottom-up design. The door slides vertically inside of specially created guide rails. The door is actuated by a servo motor through a crank-mechanism. The mechanical parts are 3D printed out of ABS to reduce the weight inside of the system.

Calculations

To make sure the chosen servo motor can actuate the door mechanism safely, the minimum torque required needs to be calculated.
Total force required: Weight of door (F_g) + Friction (F_f)

Weight: Assuming the weight is around 0.5kg for a fully filled out, printed door, the weight force will be - 0.5kg x 9.81 m/s^2 = 4.9 N.
Friction: Assuming a squirrel can carry weights of up to 300g, which equates to 0.3kg x 9.81 m/s^2 = 2.94 N on the door and further wear, crushed seed etc. causing a coefficient of friction assumed to be around 0.35, this force would be - F_f = 2.94N x 0.35 = 1 N.
So the total force needed to be lifted is F_t = 5.9 N.

Assuming the distance from the motor to the door linkage is about 3cm (0.03m), the torque required to be exerted by the motor is τ = 5.9 N x 0.03 m = 0,177 Nm = 1.8 kg*cm which gives the selected motor a safety factor of 7.2. This will ensure safe operation even with heavier obstructions.

Strength analysis

Mechanical Properties	Metric
Tensile Strength	27.6 - 34.5 MPa
Modulus of Rupture	0.0483 - 0.0689 GPa
Flexural Modulus	8.20 - 10.3 GPa
Compressive Strength	31.0 - 41.4 MPa
Shear Modulus	0.138 - 0.207 GPa
	0.586 - 0.758 GPa
Shear Strength	1.72 - 2.07 MPa
	5.52 - 6.89 MPa

Figure 1.6 - material properties of plywood

The structural analysis needs to confirm whether the assembled construction will experience failure.

Load cell: As the project constraints a strain-gauge marked for 5 kg max, the housing along with any possible intruders must not exceed a force limit of approximately 49N.

Pin-joint Shear strength: Calculations concerning the small pin connections from the servo to the links. The most probable failure would be a tear through the material caused by shear stresses from the door/motor.

Shear stress formula: $\tau = \frac{F}{A} = \frac{5.9}{3mm*2mm}$=1 MPa

With the shear strength of ABS being equal to 67 MPa, the safety factor is 67x.

Flexural strength of the door: If an intruder pushes through the door, it must be able to withstand the force applied to it. In this case the door acts as a beam supported on two sides

$$\sigma = \frac{M_{max}}{W} = \frac{\frac{F_h * L}{4}}{\frac{H*t^2}{6}}$$

F_h=Force from animal - assume 15N

L=The span of the door

H=Height of door

t=thickness of the door panel

$$\frac{\frac{15*190}{4}}{\frac{100*2^2}{6}} = 10{,}688 \text{MPa}$$

With the average flexural yield strength of ABS being equal to 72.7 MPa, this gives us a safety factor of approximately 6.8.

2. Model design

2.1. Parts list



Figure 2.11 - Arduino Uno



Figure 2.12 - RTC DS3231

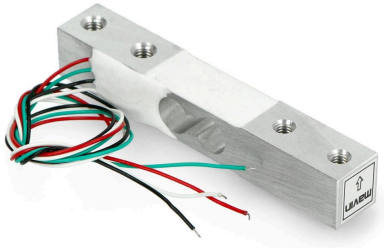


Figure 2.13 - Load Cell (5kg)

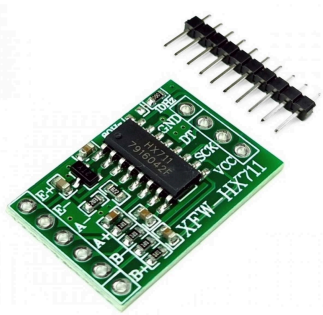


Figure 2.14 - HX711 mini module



Figure 2.15 - MG996R Servo Motor

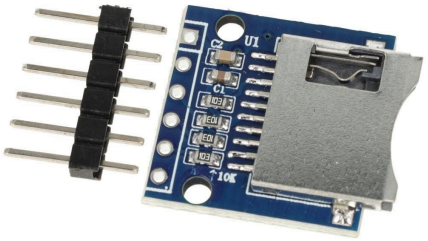


Figure 2.16 - Micro SD Card reader

2.2. Model screenshots

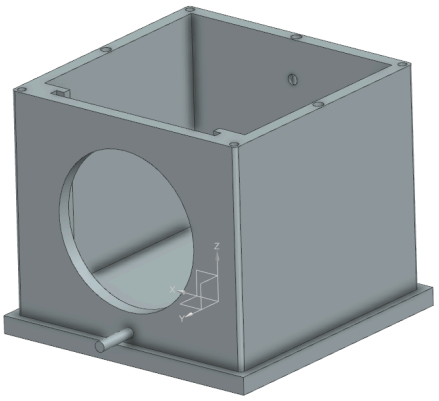


Figure 2.21 - Main Body (Part where grain and water is located)

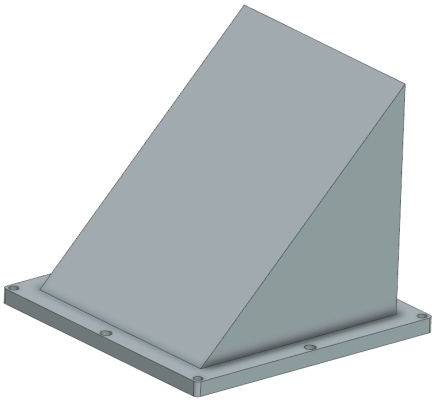


Figure 2.22 - Roof for solar panel

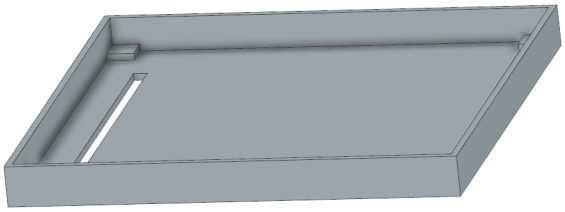


Figure 2.23 - Floor for Load Cell/Cells

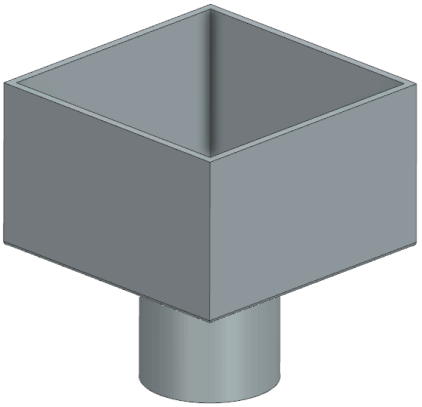


Figure 2.24 - Box for electronics and door

2.3. Assembly

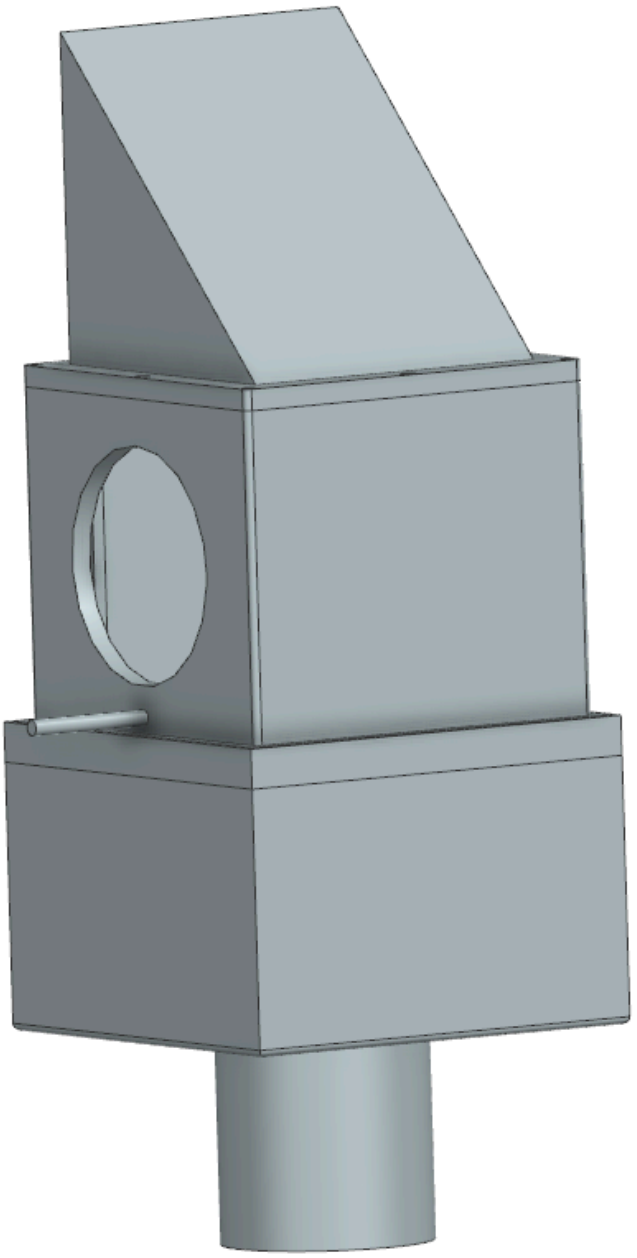


Figure 2.3 - Assembly

3. Electronic design

3.1. Design process

For the prototype, an Arduino Uno R3 microcontroller was picked due to its low cost and good availability.
The most crucial part of the project is the strain gauge - a standard 5kg load cell was chosen (for the prototype) in pair with an HX711 amplifier.
The MG996R servo was picked as an actuator due to its low power consumption and high torque. It should be noted that this is not an optimal choice for the final project as this is a 360° servo, lacking position tracking.
A standard micro SD module was picked to handle data logging, however, since Arduino Uno does not possess inherent time tracking, it was necessary to also implement a DS3231 RTC module.
The final part list for the prototype can be seen in table 3.1 under the consumption analysis section (3.3).

3.2. Electronic scheme

Once all parts were chosen, they were connected according to the following scheme:

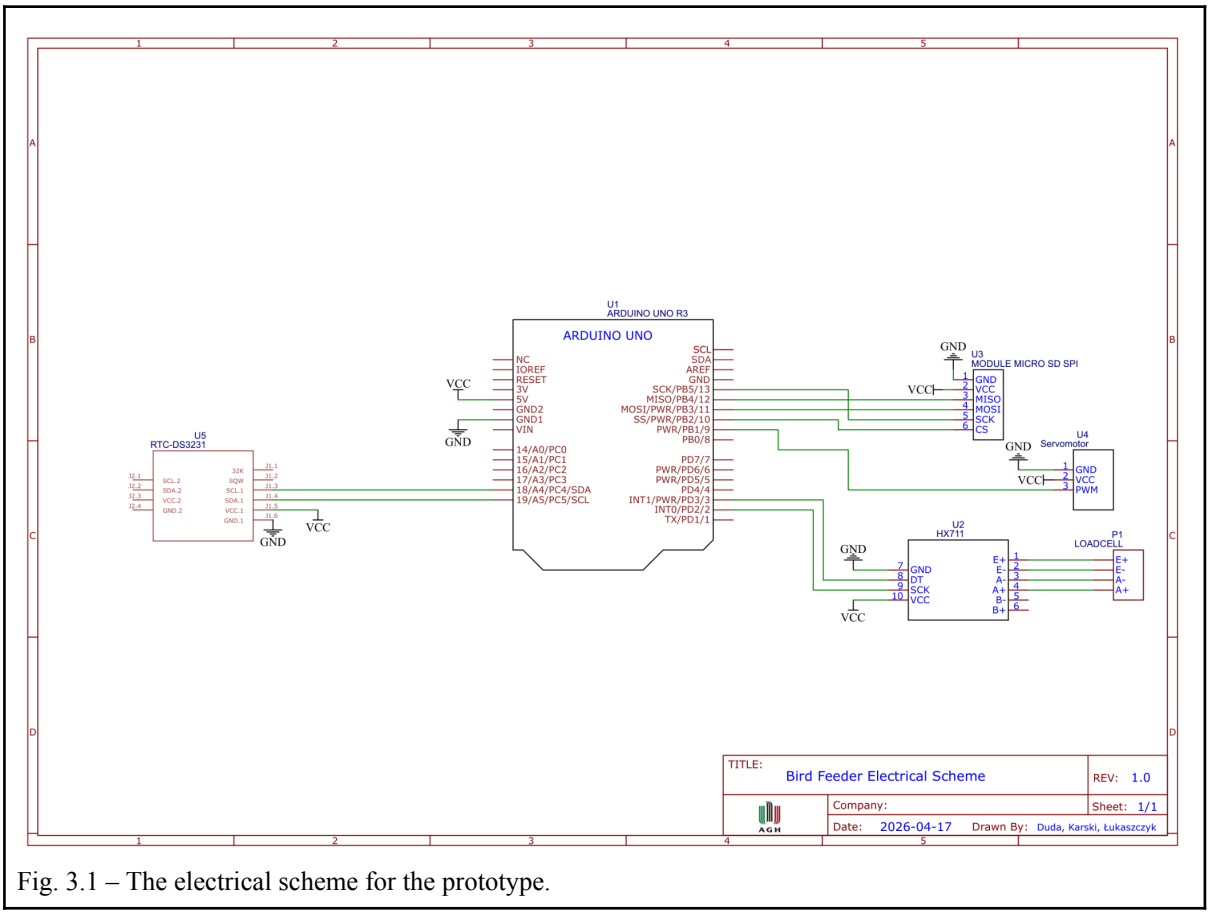


Fig. 3.1 – The electrical scheme for the prototype.

3.3. Energy consumption analysis

3.3.1. Component power draw breakdown

Component	Idle current (mA)	Peak current (mA)	Function
Arduino UNO R3	50.0	50.0	Microcontroller
DS3231 RTC	3.0	3.0	Time keeping
HX711 + load cell	15.0	15.0	Weight sensing
Micro SD module	10.0	150.0	Data logging
MG996R servo	20.0	60.0 (holding) 250.0 (moving)	Actuation
Total baseline	**~ 98.0**		

Table 3.1 – Final parts list for the prototype.

3.3.2. Operational state analysis

The system will operate entirely on a 5V DC architecture. Therefore, power consumption can be calculated with the formula:

$$P = V \cdot I$$

To calculate the average daily energy usage, the system's behaviour should be considered in three distinct states:

- **State A: Standby (idle)**
  - The system is waiting for a mass to be registered. The servo is resting.
  - Current: ~98 mA
  - Power: **0.49 W**
- **State B: Data logging (bird detected)**
  - Mass below threshold (100g) is detected. SD card write sequence is initiated.
  - Current: ~238 mA
  - Power: **1.19 W**
- **State C: Actuation (intruder detected)**
  - Mass over threshold (100g) is detected. The door closes and actively holds until the intruder leaves or until a 5-minute safety timeout is reached.
  - Duration: assumed average of 120s per event.
  - Current: ~158 mA
  - Power: ~**0.79 W** (average across movement and holding)

3.3.3. Estimated daily energy consumption

Assuming a highly active testing environment with 100 bird visits and 10 intruder events per day, with the average intruder lingering for 120s:

- **Active time:** 100s (logging) + 1200s (actuation) = 1300s = 0.36h
- **Standby time:** 85100s = 23.64h
- **Energy calculation:**
  - Standby energy: $0.49\,W \cdot 23.64\,h = 11.58\,Wh$
  - Logging energy: $1.19\,W \cdot \frac{100}{3600}\,h = 0.03\,Wh$
  - Actuation energy: $0.79\,W \cdot \frac{1200}{3600}\,h = 0.26\,Wh$
- **Total daily energy:** ~11.87 Wh

3.3.4. Cost and battery lifecycle

Two cases should be considered. For a simple prototype a powerbank would be the most efficient option.
However, for long term sustainable use, a 5W, 5V Polycrystalline solar panel is proposed, with a custom battery pack as a backup source, consisting of high-capacity 18650 lithium-ion cells wired in parallel, managed by a dedicated TP4056 charge controller and a 5V boost module:

- Powerbank (prototype only):

  Annual costs:
  11.87 Wh/day equates to 4.33 kWh per year. Assuming an average Polish electricity tariff is 1.20 PLN/kWh, the total annual cost is **5.20 PLN**.

  Battery operation:
  A standard 10000 mAh USB power bank holds roughly 31.4 Wh of usable energy (accounting for conversion losses). Thus, the estimated runtime is about **2.64 days**.

- Solar panel + battery pack:

  Solar panel energy yield:
  Assuming an average of 4 hours of direct sunlight daily and accounting for 30% inefficiency the average yield would be about:

  $$5\,W \cdot 4\,h \cdot 0.7 = 14\,\frac{Wh}{day}$$

  Assuming the total daily energy is about 11.87 Wh, the yield generates a surplus of **2.13 Wh/day**.

Since the system generates more power than it consumes, it is technically capable of infinite autonomy.
Taking into account unforeseen weather conditions, the custom battery pack should generate energy comparable to a 10000 mAh powerbank, so the system could be sustained for about 2.64 days even in complete darkness.

3.3.5. Conclusions

Even assuming a very intense environment, the system should be sustained by a small solar panel. Since poor weather conditions would likely mean less bird traffic, the battery pack alone should keep the Feeder running for about 3 days.

This is all assuming the use of an Arduino UNO development board, which is far from an optimal choice for a long-term sustainable mechanism. Over 90% of the battery's power would be wasted on the board's LED light and USB communication chip. The board was chosen due to its simplicity and cheap price but it would only be used in the prototype.

For a more sustainable system, an Arduino Pro Mini would be recommended, cutting the idle drain from 50mA to merely 5mA, meaning the same battery could sustain the system for over three weeks.

Despite all that, the conclusion stands that even assuming a suboptimal microcontroller, very intense environment and harsh weather conditions, the Bird Feeder should be able to work throughout the entire spring and summer season without requiring a recharging.

4. Software design

4.1. Design process

Since the prototype is meant to work on an Arduino Uno, the standard Arduino IDE was chosen and C++ used as the programming language.
First, a flow chart diagram (fig. 4.1) was drawn to take all requirements into account. Since many programmable components were required (like the servo, SD card reader, RTC, etc.), each of them was checked in a separate bit of code and then they were all connected in the final version (section 4.3).

4.2. Flow chart

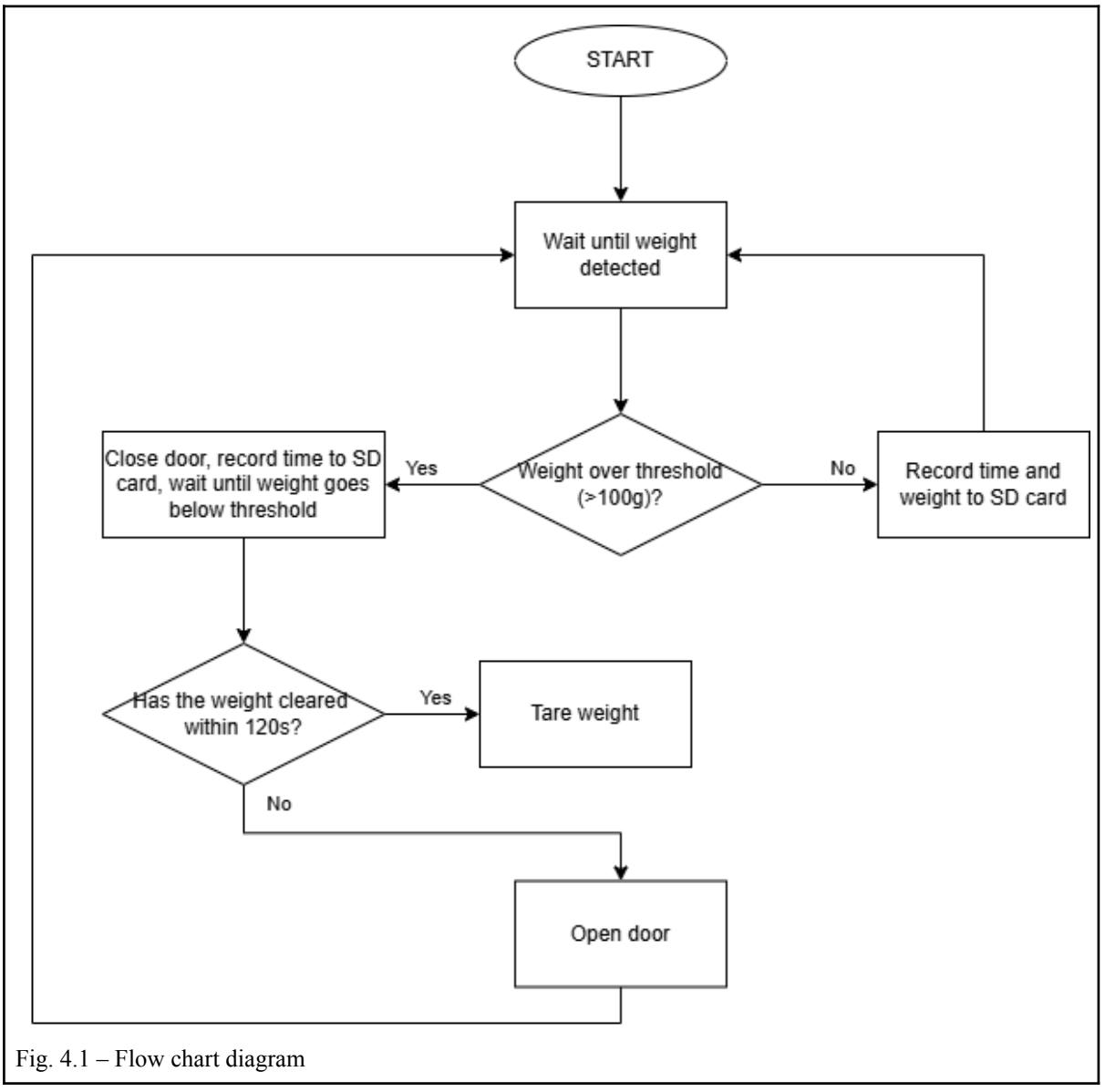


Fig. 4.1 – Flow chart diagram

4.3. Code

The final code, concluding all requirements for the prototype. It takes 800g instead of 100g as the threshold weight to make it more controllable during the presentation.

```
#include "HX711.h"
#include <Servo.h>
#include <SPI.h>
#include <SD.h>
#include "RTClib.h"

HX711 scale;
Servo doorServo;
RTC_DS3231 rtc;

uint8_t dataPin = 3;
uint8_t clockPin = 2;
const int SERVO_PIN = 9;
const int CS_PIN = 10;

const int SERVO_STOP = 90;
const int SERVO_CLOSE_SPEED = 102;
const int SERVO_OPEN_SPEED = 85;

bool isDoorClosed = false;
unsigned long lastPrintTime = 0;
unsigned long doorClosedTimestamp = 0;

void setup() {
  Serial.begin(115200);

  // 1. Initialise Servo
  doorServo.attach(SERVO_PIN);
  doorServo.write(SERVO_STOP);
  delay(200);
  doorServo.detach();

  // 2. Initialise RTC
  Serial.println(F("\nInitialising RTC..."));
  if (!rtc.begin()) {
    Serial.println(F("ERROR: Couldn't find RTC module."));
  } else {
```

```
    if (rtc.lostPower()) {
      Serial.println(F("RTC lost power, resetting time!"));
      rtc.adjust(DateTime(F(__DATE__), F(__TIME__)));
    }
  }

  // 3. Initialise SD Card
  Serial.println(F("Initialising SD Card..."));
  if (!SD.begin(CS_PIN)) {
    Serial.println(F("ERROR: SD Card failed or not present!"));
  } else {
    Serial.println(F("SUCCESS: SD Card ready."));
  }

  // 4. Initialise Load Cell
  Serial.println(F("\nInitialising Scale..."));
  scale.begin(dataPin, clockPin);

  Serial.println(F("\nEmpty the scale, press a key and hit Enter"));
  while(!Serial.available());
  while(Serial.available()) Serial.read();

  scale.tare(20);
  Serial.print(F("Tared. Current UNITS: "));
  Serial.println(scale.get_units(10));

  Serial.println(F("\nPut exactly 500g on the scale, press a key and
hit Enter"));
  while(!Serial.available());
  while(Serial.available()) Serial.read();

  scale.calibrate_scale(500, 5);
  Serial.print(F("Calibrated. Current UNITS: "));
  Serial.println(scale.get_units(10));

  Serial.println(F("\n--- SYSTEM ARMED ---"));
  lastPrintTime = millis();
}

void logEventToSD() {
  DateTime now = rtc.now();
  File dataFile = SD.open("LOG.TXT", FILE_WRITE);
```

```
  if (dataFile) {
    dataFile.print(now.year(), DEC);
    dataFile.print('/');
    dataFile.print(now.month(), DEC);
    dataFile.print('/');
    dataFile.print(now.day(), DEC);
    dataFile.print(F(" "));
    dataFile.print(now.hour(), DEC);
    dataFile.print(':');
    dataFile.print(now.minute(), DEC);
    dataFile.print(':');
    dataFile.print(now.second(), DEC);
    dataFile.println(F(" - Intruder Detected!"));
    dataFile.close();
    Serial.println(F(">>> Time logged to SD Card <<<"));
  } else {
    Serial.println(F(">>> ERROR: Could not open LOG.TXT <<<"));
  }
}

void loop() {
  float currentWeight = scale.get_units(2);

  if (currentWeight < -20.0) {
    Serial.print(F("\n>>> Weight removed ("));
    Serial.print(currentWeight);
    Serial.println(F("g). Re-taring to zero... <<<"));

    scale.tare(10);
    currentWeight = 0.0;
  }

  if (millis() - lastPrintTime >= 5000) {
    Serial.print(F("WEIGHT: "));
    Serial.print(currentWeight);
    Serial.print(F(" g   |   STATUS: "));

    if (isDoorClosed) {
      Serial.println(F("CLOSED"));
    } else {
      Serial.println(F("OPEN"));
```

```
    }
    lastPrintTime = millis();
  }


  if (currentWeight > 800.0 && isDoorClosed == false) {
    Serial.println(F("\n>>> Intruder detected! Closing... <<<"));

    doorServo.attach(SERVO_PIN);
    doorServo.write(SERVO_CLOSE_SPEED);
    delay(1500);
    doorServo.write(SERVO_STOP);
    delay(100);
    doorServo.detach();

    isDoorClosed = true;
    doorClosedTimestamp = millis();

    logEventToSD();
    Serial.println(F(">>> Door CLOSED. <<<\n"));
  }

  else if (currentWeight <= 800.0 && isDoorClosed == true) {
    Serial.println(F("\n>>> Weight cleared! Opening... <<<"));

    doorServo.attach(SERVO_PIN);
    doorServo.write(SERVO_OPEN_SPEED);
    delay(1500);
    doorServo.write(SERVO_STOP);
    delay(100);
    doorServo.detach();

    isDoorClosed = false;
    Serial.println(F(">>> Door OPEN. <<<\n"));
  }

  if (isDoorClosed == true && (millis() - doorClosedTimestamp >=
15000)) {
    Serial.println(F("\n>>> 15 SEC TIMEOUT: Forcing open! <<<"));

    doorServo.attach(SERVO_PIN);
    doorServo.write(SERVO_OPEN_SPEED);
```

```
    delay(1500);
    doorServo.write(SERVO_STOP);
    delay(100);
    doorServo.detach();

    isDoorClosed = false;

    Serial.println(F(">>> Taring scale to new baseline weight...
<<<"));
    scale.tare(10);

    Serial.println(F(">>> Door OPEN. System Reset. <<<\n"));
  }
}
```

5. Summary

The mechatronic engineering design process presented in this report successfully establishes a framework for an autonomous and energy-independent bird feeder. By utilizing a strict morphological selection matrix, the ideal combination of parts was determined, specifically combining an electronic strain-gauge with a continuous rotation servo motor and a bottom-up sliding door mechanism.

From a mechanical standpoint, structural calculations confirm that the system will operate well within safety margins. The chosen MG996R servo motor produces an abundant safety factor of 7.2 when overcoming worst case gravitational and frictional forces during door actuation. Furthermore, analytical stress evaluations prove that both the pin-joints and the 3D-printed ABS sliding panel can effortlessly withstand maximum simulated intruder induced forces, yielding safety margins of 6.8 against flexural deformation and over 60 against material shear failures.

On the electrical side, power-consumption state modeling reveals a baseline daily operational requirement of 11.87 Wh under intensive use conditions. This demand is easily countered by the 5V solar panel, which yields an expected average of 14 Wh per day, resulting in a consistent net energy surplus of 2.13 Wh/day. Therefore, the target device achieves complete operational autonomy, with the underlying lithium-ion reserve capacity sustaining regular behaviour through even 2.64 days of continuous darkness.

Further upgrades to the project may include the use of a lower-power servo motor/linear actuator, a lower-power microcontroller and possibly employing plastic for the housing of the feeder.

6. Attachments

Presentations
[1] Initial presentation
[2] Midterm presentation
[3] Final presentation

Models
[4] eleBox.prt – Box for electrical components
[5] floor&doorhole&cable.prt – Main floor
[6] Main – Main body
[7] Roof.prt – Roof with solar panel
[8] Tensometer_Place.prt – Floor with the load cell